## STAINLESS STEEL



▲ Figure 15.13 Some cars have a stainless steel body. They do not need to be painted.

Stainless steel is an alloy of iron with chromium and often nickel. Chromium forms a strong oxide layer in the same way as aluminium, and this oxide layer protects the iron as well. Stainless steel is therefore very resistant to corrosion.

Uses include kitchen sinks, saucepans, knives and forks, and gardening tools, but there are also major uses for it in the brewing (making beer), dairy (milk and cheese production) and chemical industries, where corrosion-resistant vessels are essential. Stainless steel is significantly more expensive than mild steel.

Table 15.1 A summary of types of steel

| Type of steel | Iron mixed with | Some uses |
|---|---|---|
| mild steel | up to 0.25% carbon | nails, car bodies, ship building, girders |
| high-carbon steel | 0.6–1.2% carbon | cutting tools, masonry nails |
| stainless steel | chromium (and nickel) | cutlery, cooking utensils, kitchen sinks |

## COPPER



▲ Figure 15.14 Copper pots and pans

Some properties and uses of copper and its alloys are summarised in Table 15.2.

Table 15.2 Copper and its alloys have a variety of uses

| Use | Property |
|---|---|
| electrical wires | very good conductor of electricity and ductile |
| pots and pans | very good conductor of heat (thermal conductor), very unreactive and malleable |
| water pipes | unreactive – does not react with hot or cold water and malleable |
| surfaces in hospitals | antimicrobial properties and malleable |

## CHAPTER QUESTIONS

SKILLS REASONING, PROBLEM SOLVING 9

1 Explain why different methods are used to extract aluminium and iron from their ores and write equations for the reactions occurring.

SKILLS CREATIVITY 10

2 Sodium is the sixth most abundant element in the Earth's crust, occurring in large quantities as common salt, NaCl, and yet sodium metal wasn't first produced until the early 19th century.

a From your knowledge of the position of sodium in the reactivity series, suggest a method for manufacturing sodium from sodium chloride. You aren't expected to give details of the manufacturing process, but should describe and explain (including equation(s) where relevant) how sodium is formed in your process.

SKILLS REASONING 7

b Suggest why sodium wasn't produced until the early 19th century.

c Suggest three other metals which might have been first isolated from their compounds at the same sort of time.

SKILLS CREATIVITY (8)

3 Lead is between iron and copper in the reactivity series. Explain a method that could be used to extract lead from its ore.

4 The extraction of chromium from its ore is quite complicated, but in the final stage chromium(III) oxide is reacted with aluminium to form chromium and aluminium oxide.

SKILLS PROBLEM SOLVING (9)

a Write a balanced equation for this reaction.

SKILLS REASONING

b Explain what the reducing agent is in this reaction.

(8)

c Explain whether chromium is more or less reactive than aluminium.

5 The first stage in the extraction of titanium from its ore is heating the ore with chlorine and carbon. The equation for the reaction is:

$$2FeTiO_3 + 7Cl_2 + C \rightarrow 2TiCl_4 + 2FeCl_3 + CO$$

This equation is not completely balanced.

The $TiCl_4$ is separated from the $FeCl_3$ and then, in the second stage, heated with magnesium.

SKILLS PROBLEM SOLVING

a Balance the equation for the first stage in this process.

(9)

b Write a balanced chemical equation for the second stage in the process.

SKILLS REASONING (8)

c Explain whether magnesium is more or less reactive than titanium.

SKILLS CREATIVITY

d Explain a method that could be used to extract magnesium from its ore.

SKILLS REASONING

e Explain whether you would expect titanium to be more or less expensive than magnesium.

SKILLS CRITICAL THINKING (4)

6 This question is about the properties and uses of some metals.

a Aluminium alloys are used in aircraft construction.

i Explain which property of aluminium makes it particularly suitable for this purpose.

SKILLS REASONING (8)

ii Explain, in terms of structure and bonding, why aluminium alloys are used in preference to pure aluminium.

(7)

iii Some cars have aluminium bodies rather than mild steel bodies. Explain two advantages and one disadvantage of using aluminium rather than mild steel.

b Explain two properties of copper that make it suitable for use in:

(4)

i electrical wiring

(5)

ii water pipes

(7)

c Explain why car bodies are made from mild steel rather than high-carbon steel.

END OF CHEMISTRY ONLY